

Freight Rate Prediction: Model Validation and Data Assessment Report

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1. Data Profile & Quality Insights

Initial exploration of the dataset, which covers a 10-month shipping timeline from January 1, 2025, to October 31, 2025, revealed critical operational patterns and minor data quality constraints. A systematic preprocessing pipeline was applied prior to modeling:

- **Missing Values Handling:** The initial dataset contained relatively few missing values, specifically 300 in the weight feature and 374 in market_index. These were successfully resolved using median imputation to maintain data integrity and robustness against outliers.
- **Physical Anomaly Correction:** A total of 292 rows contained anomalous negative weight values (e.g., -36559.0). Because negative physical weight is an impossibility, this entry error was corrected by converting the values to their absolute equivalents.
- **Equipment Profile:** Analysis confirmed that the shipments within this dataset utilize three primary equipment types: Dry Van, Reefer, and Flatbed. These were categorically encoded for algorithm compatibility.

2. Pricing & Rate Per Mile Observations

- **Deterministic Correlation:** A highly significant linear relationship (Pearson correlation coefficient of 0.91) exists between shipment distance and the target variable posted_rate. This establishes distance as the overwhelming primary driver behind freight pricing.
- **Distinct Pricing Tiers:** Visualizing distance versus posted rate revealed three distinct linear pricing bands. When segmented by equipment and quote_signal, it became clear that different vehicle configurations and underlying signaling structures operate under different established rate schedules.
- **Rate per Mile Metrics:** The average rate per mile across the data is **\$2.22** (median **\$2.15**), with the vast majority of shipments clustered tightly in the **\$1.80 to \$2.80** range. High outliers in the rate-per-mile distribution consistently correspond to very short-haul trips, which naturally command a premium relative to distance.

3. Validation Strategy & Model Performance

To construct a model capable of genuine forecasting, a chronological data split was implemented. By indexing the timeline, the model was trained exclusively on earlier historical timeframes and validated against a chronologically later hold-out set, accurately mirroring the real-world constraints of predicting future rates without data leakage.

Algorithm Evaluation:

An initial Random Forest Regressor was trained to establish a performance baseline. The architecture was then transitioned to a LightGBM Regressor, which efficiently captured the tiered pricing dynamics and improved overall predictive accuracy.

Model Architecture	R2 Score	Mean Absolute Error (MAE)
Random Forest (Baseline)	0.8477	\$131.01
LightGBM (Champion)	0.8645	\$120.51

Key Predictors:

Based on the LightGBM feature importances, the top four variables driving shipment pricing are:

- 1. distance (Most important by a significant margin)
- 2. quote_signal
- 3. market_index
- 4. weight

4. Downstream Applications & Forecasting

- **Validation Predictions:** The champion LightGBM model was successfully applied to generate final predictions on the 12,000-row hold-out validation set (validation_predictions.csv), demonstrating its capability for bulk production scoring.
- **Operational Forecasts (December 2025):** The pipeline was additionally utilized to predict pricing on a custom monthly scenario (december-chart-inputs.csv) to facilitate capacity and forward planning for December 2025. The model accurately captured the expected weekly cyclical rate fluctuations.

